MODULE 21 · SECTION 3 - STORING THE DATA

Relational and Operational Stores

Cloud SQL availability, replicas and connections; Spanner keys, interleaving, configurations and transactions; Memorystore tiers.

SECTION 3 of 5 EXAM WEIGHT 20% MODULE 21 of 45

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High availability keeps you up. Replicas spread the reads. **Only some stores spread the writes.**

High availability or read replica?

HIGH AVAILABILITY

Stay up when a zone fails

- Primary and standby in two zones
- Synchronous, so no committed data lost
- Failover in about 60 seconds
- Standby does not serve reads

READ REPLICA

Spread the reads

- Asynchronous copy, slightly behind
- Read-only, offloads reports
- Does not fail over by itself
- Ten or fewer direct replicas advised

THE TIEBREAKER

'Survive a zone failure automatically' is HA. 'Reports slow the primary' is a read replica.

Surviving a regional outage

- A **cross-region read replica** keeps a copy in another region.
- In a disaster you **promote** it, and it becomes a standalone primary.
- Replication is asynchronous, so the last moments of writes may be lost.
- **Enterprise Plus** adds advanced disaster recovery and a **write endpoint**.

> REMEMBER
>
> HA covers a zone. A cross-region replica covers a region. Many designs need both.

Connecting safely

- Google recommends **private IP**: private services access, or Private Service Connect across VPCs.
- On public IP, use the **Cloud SQL Auth Proxy** or a **Language Connector**.
- **IAM database authentication**: short-lived tokens instead of passwords.
- Direct private connections are fastest, but you **manage SSL** yourself.

> THE NUMBERS
>
> Instances hold up to 64 TB on dedicated cores. Past that, the workload is outgrowing Cloud SQL.

Larkspur's tariff database

```
gcloud sql instances create lk-tariffs-prod \
    --database-version=POSTGRES_16 \
    --edition=ENTERPRISE_PLUS --tier=db-perf-optimized-N-8 \
    --region=europe-west2 --availability-type=REGIONAL \
    --network=projects/lk-network-prod/global/networks/lk-vpc \
    --no-assign-ip \
    --database-flags=cloudsql.iam_authentication=on

# A cross-region replica for disaster recovery
gcloud sql instances create lk-tariffs-dr \
    --master-instance-name=lk-tariffs-prod --region=europe-west1
```

4 Regional means high availability: a standby in a second zone.

6-7 Private IP only, and IAM logins instead of passwords.

CHECK RESIDENCY

Tariffs carry no residency rule, so a replica outside the UK is allowed. German personal data could not do this.

Spanner keys: spread the writes

- Never lead a busy table's key with a value that **only goes up**, or only down.
- Use **UUID version 4**. Version 1 hides a timestamp at the front.
- For numbers, a **bit-reversed sequence**. Or swap the key order, identity first.
- Indexes are tables too, so the **same rule** applies to indexed columns.

> THE TRAP
>
> Reversing the sort order does not help. A descending sequence is still monotonic.

Keys and interleaving, if billing moved to Spanner

```
 1  CREATE TABLE Accounts (
 2    AccountId  STRING(36) NOT NULL,   -- UUID v4, never a sequence
 3    CustomerId STRING(36) NOT NULL,
 4    TariffId   STRING(20),
 5  ) PRIMARY KEY (AccountId);
 6
 7  CREATE TABLE Bills (
 8    AccountId  STRING(36) NOT NULL,
 9    BillDate   DATE NOT NULL,
10    AmountGbp  NUMERIC,
11  ) PRIMARY KEY (AccountId, BillDate DESC),
12    INTERLEAVE IN PARENT Accounts ON DELETE CASCADE;
```

2 A random key spreads new accounts across every split.

11-12 Bills live beside their account, newest first.

THE PAYOFF

Reading an account and its recent bills becomes one local read, not a join across servers.

Spanner configurations

CONFIGURATION	REPLICAS	AVAILABILITY
Regional	3 read-write, in three zones	99.99%
Dual-region	6, across two regions in one country	99.999%
Multi-region	Two read-write regions, a witness, optional read-only	99.999%

THE TRADE-OFF

Writes need a quorum, so multi-region writes cost a little latency. Reads do not need a quorum.

Read-write or read-only transactions?

READ-WRITE

For changes

- Takes locks
- May abort, so retry
- Keep it short
- Externally consistent

READ-ONLY

For reading

- No locks, blocks nobody
- A consistent snapshot
- Can read slightly in the past
- Lower latency, fewer aborts

THE RULE

Use read-only for reads. For bulk updates, Partitioned DML; for writes that need no atomicity, batch writes.

Memorystore tiers and failover

- **Redis Basic**: one node, no replica, no failover. A pure cache.
- **Redis Standard**: a replica in another zone, with **automatic failover**.
- **Read replicas** scale reads. Redis Cluster and Valkey cluster mode shard data.
- Tiers cannot be converted. Move with **export and import**.

> THE SIGNAL
>
> 'The cache must survive a node or zone failure' rules out Basic.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Survive a **zone** failure automatically	Cloud SQL high availability
Reports slow the primary	A read replica
Survive a **region** failure	Cross-region replica, promote in a disaster
Connect **without passwords**	Auth Proxy or connector, IAM database auth
Spanner writes on **one split**	UUID v4 or bit-reversed keys
Dashboard reads that **must not block**	Spanner read-only transactions

THE PATTERN

Up, read, write: three different dials. Find which one the question is turning.

> Larkspur's tariff database runs on a single-zone Cloud SQL instance in europe-west2.
>
> If that zone fails, the database must come back automatically within minutes, without losing any committed change.

What should the team configure?

A. High availability, as a regional instance with a standby

B. A read replica placed in a different zone of the same region

C. Nightly exports of the database to a Cloud Storage bucket

D. A cross-region replica in europe-west1, promoted by hand

◆ PAUSE AND DECIDE

What should the team configure?

A High availability

Synchronous standby in another zone, automatic failover in about a minute.

B A read replica

Asynchronous, and it does not fail over by itself.

C Nightly exports

Up to a day of lost changes, and a slow manual restore.

D A cross-region replica

Manual promotion, asynchronous, and a region is not the risk here.

THE SIGNAL

Zone failure, automatic, no committed data lost: Cloud SQL high availability.

> Larkspur's pricing analysts run heavy reports against the tariff database every morning, and the application slows down while they run.
>
> The reports can use data that is a few seconds old.

What should the team do?

A Turn on high availability, and send the reports to the standby

B Create a read replica, and point the reporting tool at it

C Double the primary's machine size before every morning run

D Cache every report's result in Memorystore for the analysts

◆ PAUSE AND DECIDE

What should the team do?

A Reports on the standby

The HA standby does not serve reads.

B A read replica

Reports run on the copy, and the primary serves the app.

C A bigger primary

Pays for a larger machine to keep two workloads fighting.

D Cache the reports

Heavy, varied reports change; a cache does not run them.

THE SIGNAL

Reads slowing the primary, a little staleness allowed: a read replica.

> A Spanner table of meter events uses EventId as its primary key, taken from a sequence that increases by one each time.
>
> Write latency climbs at peak, and monitoring shows most writes landing on a single split.

What should the team change?

A Declare the key in descending order, so new rows sort first

B Add nodes to the instance, so that the hot split has more capacity

C Key the table with UUID v4 values, or a bit-reversed sequence

D Add a secondary index on EventId, to spread the reads

◆ PAUSE AND DECIDE

What should the team change?

A **Descending order**

Still monotonic, so writes still pile onto one end.

B **More nodes**

One hot split stays on one server, however many there are.

C **UUID v4 or bit-reversed**

Random-looking keys spread writes across every split.

D **An index on EventId**

Indexes are tables, so it hotspots too, and reads were not the issue.

THE SIGNAL

Writes on one split from a sequence key: make the key non-monotonic.

A Larkspur service on GKE must connect to a Cloud SQL instance that currently has only a public IP address.

Security wants encrypted connections, no long-lived database passwords, and no certificates to manage by hand.

How should the service connect?

A Add 0.0.0.0/0 to authorized networks, and use a strong password

B Connect directly, with hand-managed SSL certificates and a shared password

C Copy the needed tables into BigQuery, and query them from there

D Use the Cloud SQL Auth Proxy, with IAM database authentication

◆ PAUSE AND DECIDE

How should the service connect?

A Open to the internet

Exposes the database to everyone, with a long-lived password.

B Hand-managed SSL

Certificates and a shared password, both of which security ruled out.

C Copy into BigQuery

Changes the architecture, and the service needs the live database.

D Auth Proxy and IAM

Encrypted, identity-checked, with short-lived tokens refreshed for you.

THE SIGNAL

Public IP, no passwords, no certificates to manage: Auth Proxy or a connector, with IAM auth.

> A dashboard reads across several Spanner tables every few seconds. Its numbers must be consistent with each other, and data up to 10 seconds old is fine.
>
> Its queries must never block or slow down the application's writes.

Which approach fits best here?

A Read-only transactions, reading at a timestamp a few seconds old

B Read-write transactions, so that every single read is fully locked

C A nightly copy of the tables in Cloud SQL, for the dashboard

D Partitioned DML statements, run by the dashboard each time

◆ PAUSE AND DECIDE

Which approach fits best here?

A **Read-only, slightly stale**
No locks, a consistent snapshot, and lower latency.

B **Read-write**
Locks and aborts, exactly what must not touch the writers.

C **A nightly Cloud SQL copy**
A day stale, when ten seconds was the limit.

D **Partitioned DML**
A tool for bulk changes, not for reading.

THE SIGNAL

Consistent reads that must not block writers: read-only transactions, optionally stale.

> Larkspur caches tariffs in a Memorystore for Redis Basic instance. When its node failed last month, every request fell through to Cloud SQL at once, and the database buckled.
>
> The cache must now survive a node or zone failure automatically.

What should the team do?

A. Resize the Basic tier instance to a much larger memory size

B. Move to Standard tier, with a replica in another zone

C. Store the cache's contents in Cloud SQL as a backup copy

D. Raise the cache entry TTLs, so that items last for days

◆ PAUSE AND DECIDE

What should the team do?

A. A larger Basic instance
Still one node, with no replica and no failover.

B. Standard tier
A replica in another zone, and automatic failover.

C. Back up into Cloud SQL
Puts the load back on the database it was protecting.

D. Longer TTLs
Staler data, and the next failure still empties everything.

THE SIGNAL

A cache that must survive a node or zone failure: Standard tier, not Basic.

Module 21 in five lines

- Cloud SQL **HA** survives a zone. **Read replicas** spread reads, and do not fail over.
- A **cross-region replica**, promoted in a disaster, covers a region.
- Private IP, the **Auth Proxy** or connectors, and **IAM database auth**.
- Spanner: no monotonic keys, **interleave** children, read-only transactions for reads.
- Memorystore **Basic** has no failover. **Standard** does.

COMING UP

Next: module 22, Cloud Storage classes, lifecycle and retention.